

Component Inventory & Storybook Implementation Specification

Version 0.1

Implementation axiom

A UI component SHALL encode reusable visual and interaction semantics, not feature-specific assumptions. Storybook SHALL make those semantics inspectable, testable, accessible, and regression-safe before components are composed into production screens.

Field	Value
Document ID	CSAML-CISB-0.1
Status	Normative frontend implementation baseline for MVP 0.1
Primary owners	Frontend Engineering / Design System Engineering
Inputs	UX, IA, Screen Inventory, Wireframe, UI Design System, High-Fidelity UI
Primary output	Reusable component library + Storybook contract + testable component documentation

1. Purpose and Scope

This specification defines the reusable CS-AML frontend component inventory and the Storybook implementation contract for MVP 0.1. It is the implementation bridge between the normative UI specifications and production React code. It defines component boundaries, ownership, variants, accessibility expectations, stories, testing, visual regression, documentation, and release criteria.

Boundary

This document does not redefine screen navigation, workflow, wireframe composition, or visual semantics. Those remain governed by IA, UX, Wireframe, UI Design System, and High-Fidelity UI specifications. This document converts those decisions into reusable frontend implementation units.

2. Objectives

- Prevent feature teams from recreating near-duplicate controls and analytical visuals.
- Make uncertainty, provenance, review state, permission state, and value-flow epistemics reusable rather than screen-specific.
- Provide a single inspectable catalogue for designers, frontend engineers, QA, accessibility review, and product review.
- Support isolated accessibility and interaction testing before components are integrated into full screens.
- Enable visual regression detection for design-system changes.
- Maintain screen-to-component traceability for critical MVP screens.

3. Component Architecture

Layer	Purpose	Dependency rule
Foundation	Tokens, typography, surfaces, icon wrapper, layout primitives	May depend only on framework/runtime and token definitions.
General UI	Reusable controls and generic states	May depend on Foundation; SHALL NOT import domain feature modules.
Analytical / Domain	CS-AML semantic components	May depend on Foundation + General UI; SHALL expose analytical semantics explicitly.
Compound Patterns	Reusable page/workspace compositions	May depend on all shared components; SHALL remain feature-agnostic enough to support multiple screens.
Feature Screens	Route-specific composition and data binding	Consumes shared components; new local primitives require design-system review.

CISB-001 — Feature modules SHALL consume shared design-system components before introducing local equivalents.

CISB-002 — A shared component SHALL NOT encode business-specific API calls or permission decisions internally unless the component is explicitly an authorization-aware system component.

CISB-003 — Analytical semantics such as Fact, Inference, Confidence, DIRECT, or RECONSTRUCTED SHALL be represented through typed component variants rather than arbitrary CSS classes.

4. Naming and Package Conventions

Concern	Convention
Component IDs	Stable inventory IDs: FND-###, GEN-###, ANA-###, CMP-###.
React names	PascalCase and semantic: EvidenceCitation, FlowClassBadge, ReviewDecisionPanel.
Files	ComponentName.tsx, ComponentName.stories.tsx, ComponentName.test.tsx, ComponentName.types.ts.
Exports	Public exports through package barrel; internal helpers remain private.
Variants	Typed unions/enums; avoid free-form string variant names.
Styling	Semantic tokens/CSS variables + shared Tailwind mappings; no feature-specific raw hex where semantic token exists.
Events	Intent-oriented names: onApprove, onResolve, onOpenEvidence; not onClick.
Data	Presentational components receive explicit view models; canonical API response shape SHOULD NOT leak directly into low-level components.

5. Component Inventory

The MVP baseline contains 65 shared components across foundation, general, analytical/domain, and compound pattern layers.

Foundation components

ID	Component	Priority	Purpose	Primary usage	Semantic owner
FND-001	Semantic Token Provider	P0	CSS variable/theme provider	All	DS tokens
FND-002	Icon	P0	Accessible icon wrapper	All	iconography
FND-003	Text	P0	Typography primitive with semantic variants	All	typography
FND-004	Stack / Inline	P0	Spacing/layout primitives	All	spacing
FND-005	Surface	P0	Canvas/panel/elevated surface	All	surface tokens

General components

ID	Component	Priority	Purpose	Primary usage	Semantic owner
GEN-001	Button	P0	Primary/secondary/quiet/destructive actions	All	action semantics
GEN-002	IconButton	P0	Compact labelled action	All	action semantics
GEN-003	Link	P0	Internal/external navigation link	All	navigation
GEN-004	TextInput	P0	Single-line text field	Forms	form controls
GEN-005	TextArea	P0	Long-form input	Editors	form controls
GEN-006	Select	P0	Controlled vocabulary selection	Forms	form controls
GEN-007	Combobox	P0	Search/select canonical object or term	Forms	findability
GEN-008	Checkbox	P0	Independent binary selection	Forms	form controls
GEN-009	RadioGroup	P0	Mutually exclusive selection	Forms	form controls

ID	Component	Priority	Purpose	Primary usage	Semantic owner
GEN-010	Switch	P1	Immediate reversible setting	Admin	form controls
GEN-011	DatePrecisionField	P0	Exact/month/year/range/unknown dates	Timeline	temporal semantics
GEN-012	CurrencyValueField	P0	Known/range/approximate/unknown values	Value Flow	value semantics
GEN-013	Badge	P0	Compact non-interactive semantic label	All	status
GEN-014	Chip	P0	Removable/filter/token item	Search/filters	filters
GEN-015	Alert	P0	Inline message with severity	All	feedback
GEN-016	Toast	P0	Transient action feedback	All	feedback
GEN-017	Modal	P0	Blocking confirmation/dialog	All	high-impact action
GEN-018	Drawer	P0	Context/inspector panel	Analytical	context
GEN-019	Tabs	P0	Local subnavigation	Object pages	local navigation
GEN-020	Breadcrumbs	P0	Context path/deep-link cue	Object pages	navigation
GEN-021	DataTable	P0	Sortable/faceted permission-aware table	Registers	data dense
GEN-022	Pagination	P0	Paged large result navigation	Registers/Search	navigation
GEN-023	FilterBar	P0	Filters/facets/query chips	Registers/Search	findability
GEN-024	EmptyState	P0	No-data/no-result guidance	All	states
GEN-025	LoadingState	P0	Skeleton/progress state	All	states
GEN-026	AccessDeniedState	P0	Non-disclosing permission denial	System	security

ID	Component	Priority	Purpose	Primary usage	Semantic owner
GEN-027	IntegrityWarningState	P0	Evidence integrity mismatch warning	Evidence	security
GEN-028	ConflictState	P0	Stale/concurrent edit conflict	Editors	versioning

Analytical components

ID	Component	Priority	Purpose	Primary usage	Semantic owner
ANA-001	AnalyticalStateBadge	P0	Claim/Fact/Inference/Candidate/Disputed/Unknown	Analysis	uncertainty
ANA-002	ConfidenceBadge	P0	Controlled confidence label + rationale affordance	Assessment	confidence
ANA-003	FlowClassBadge	P0	DIRECT/DOCUMENTED/RECONSTRUCTED/HYPOTHETICAL	Value Flow	epistemic class
ANA-004	EvidenceCitation	P0	Canonical evidence citation with location	Evidence/Product	provenance
ANA-005	ProvenanceTrail	P0	Source→Evidence→Derivative lineage summary	Evidence	provenance
ANA-006	SourceRating	P0	A-F reliability display/input	Source	source evaluation
ANA-007	CredibilityRating	P0	1-6 information credibility	Claims	information evaluation
ANA-008	EntitySummaryCard	P0	Canonical identity summary without accusation	Entity	identity
ANA-009	RelationshipSummary	P0	Typed evidence-backed relationship summary	Entity/Graph	relationship

ID	Component	Priority	Purpose	Primary usage	Semantic owner
ANA-010	AssetAttribution	P0	Ownership/control/use attribution display	Asset	asset attribution
ANA-011	TemporalPrecision	P0	Exact/approx/range/unknown date cue	Timeline	temporal semantics
ANA-012	EvidenceInspector	P0	Evidence/provenance/context rail	Evidence/Analysis	provenance
ANA-013	GraphNode	P0	Neutral canonical-object node	Graph	graph semantics
ANA-014	GraphEdge	P0	Evidence-backed relationship/value-flow edge	Graph	graph semantics
ANA-015	GraphLegend	P0	Visible relationship/flow semantics	Graph	legend
ANA-016	TimelineItem	P0	Evidence-linked event item	Timeline	chronology
ANA-017	HypothesisCard	P0	Hypothesis state, rationale and evidence balance	Hypothesis	reasoning
ANA-018	EvidenceBalanceMatrix	P0	Support/contradict/neutral/unknown matrix	Hypothesis	reasoning
ANA-019	IntelligenceGapItem	P0	Explicit unknown/gap display	Assessment	uncertainty
ANA-020	TypologyMatchPanel	P0	Indicators/counter-indicators/alternatives	Typology	typology caution
ANA-021	AssessmentSummary	P0	Judgement, confidence, basis, gaps	Assessment	assessment
ANA-022	ReviewDecisionPanel	P0	Comment/request changes/approve/reject	Review	review

ID	Component	Priority	Purpose	Primary usage	Semantic owner
ANA-023	ClassificationBanner	P0	Handling/ classification context	All sensitive	security
ANA-024	ProtectedSource Marker	P0	Authorized protected-source presence indicator without identity leak	Source	source protection

Compound components

ID	Component	Priority	Purpose	Primary usage	Semantic owner
CMP-001	RegisterPageFrame	P0	Header + filters + DataTable + pagination	Registers	WF-PAT-01
CMP-002	CanonicalObjectHeader	P0	Object identity/status/classification/actions	Object detail	WF-PAT-02
CMP-003	ObjectDetailFrame	P0	Context header + tabs + main + inspector	Object detail	WF-PAT-02
CMP-004	EditorFrame	P0	Form sections + validation + action footer	Editors	WF-PAT-03
CMP-005	AnalyticalWorkspaceFrame	P0	Toolbar + canvas + inspector + accessible alternative	Analysis	WF-PAT-04
CMP-006	CompareResolutionFrame	P0	Symmetric A/B comparison + differences + decision	Entity match	WF-PAT-05
CMP-007	ReviewApprovalFrame	P0	Artifact + review panel + decision controls	Review	WF-PAT-06
CMP-008	SystemStateFrame	P0	Safe failure/permission/integrity state	System	WF-PAT-07

6. Component API Contract

CISB-010 — All public components SHALL define explicit TypeScript props and SHALL avoid `any` in public interfaces.

CISB-011 — Status and analytical variants SHALL use controlled typed values aligned to normative vocabularies.

CISB-012 — Components that render canonical objects SHOULD accept stable IDs/links separately from display labels so canonical navigation remains explicit.

CISB-013 — Components rendering restricted content SHALL support redacted/withheld states without leaking hidden metadata through tooltips, DOM labels, or story fixtures.

CISB-014 — Destructive or high-impact actions SHALL expose disabled, pending, success, failure, and confirmation states where applicable.

CISB-015 — Components SHALL permit accessible names independent from visible iconography or abbreviated labels.

CISB-016 — Loading state SHALL not erase surrounding context needed to understand which object is loading.

7. Accessibility Contract

Component class	Mandatory accessibility behavior
Buttons/actions	Keyboard operable; visible focus; accessible name; disabled vs aria-disabled used intentionally.
Dialogs/drawers	Focus trap where blocking; focus restoration; labelled title; Escape behavior where safe.
Tables	Semantic headers; keyboard/screen-reader discoverable sort state; no permission leak in hidden rows.
Tabs	ARIA tab semantics or equivalent; roving focus; current state announced.
Forms	Programmatic label, help, error association; required state not color-only.
Graph/timeline/value flow	Accessible alternative list/table preserving equivalent analytical semantics.
Analytical states	Color-independent icon/pattern/text label for Claim/Fact/Inference etc.
Flow classes	Line pattern + text/legend; not color-only.
Toast/alerts	Appropriate live-region priority; avoid repeated noisy announcements.
Protected content	Accessible text SHALL not reveal information withheld visually.

CISB-020 — Storybook accessibility checks SHALL be required for P0 interactive shared components.

8. Storybook Architecture

Storybook purpose

Storybook is the executable catalogue of component semantics. It SHALL serve design review, engineering documentation, accessibility review, interaction testing, visual regression, and reference composition. It is not merely a gallery of default states.

Storybook group	Coverage	Required story emphasis
Foundations	Tokens, typography, surfaces, icons	Docs only + smoke stories
General/Actions	Button, IconButton, Link	Variants, disabled, loading, keyboard
General/Forms	Inputs, select, combobox, dates, values	Default, error, disabled, readonly, long content
General/Data Display	Badge, chip, table, pagination	Dense, empty, long values, permission-filtered
General/Feedback	Alert, toast, modal, drawer	Severity, focus trap, dismissal, async action
Analytical/Provenance	Citation, provenance, evidence inspector	Complete/missing/restricted/derivative
Analytical/Uncertainty	State badge, confidence, gaps	All epistemic states + accessible labels
Analytical/Entity	Entity card, relationship, asset attribution	Normal/conflicting/disputed/unknown
Analytical/Value Flow	Flow badge, edge, legend, value field	All four classes, grayscale, long chain
Analytical/Reasoning	Hypothesis, evidence matrix, typology, assessment	Support/contradiction/unknown/reviewed
Analytical/Review	Review panel, classification, protected-source marker	Pending/change/approved/rejected/restricted
Patterns	Register/Object/Editor/Workspace/Compare/Review/System	Reference composition stories

9. Required Story Set Per Component

Story	Requirement
Default	Representative normal state with realistic but synthetic safe fixture data.
Variants	Every public semantic variant, including all analytical states.
Interactive	User-visible interaction tested via play function where meaningful.
Keyboard	Focus order and keyboard operation demonstrated for complex interactive components.
Error / invalid	Validation or failure state where applicable.
Loading / pending	Async state where applicable.
Empty / unknown	Unknown and missing data distinct from zero/false.
Readonly / approved	Immutable/read-only state where applicable.
Restricted	Permission-safe withheld state for security-sensitive components.
Long content	Long names, identifiers, citations, translations, and wrapped labels.
Narrow viewport	Component behavior under constrained width.
High contrast / grayscale	Critical analytical semantics remain understandable without normal color perception.

CISB-030 — A P0 component is not Storybook-complete if only a default happy-path story exists.

CISB-031 — Fixtures SHALL be synthetic and SHALL NOT include protected-source identities or copied sensitive production evidence.

10. Critical Analytical Story Matrices

AnalyticalStateBadge

- Claim
- Fact
- Inference
- Candidate
- Disputed
- Unknown
- With icon hidden (a11y name preserved)
- Grayscale

FlowClassBadge / GraphEdge

- DIRECT
- DOCUMENTED
- RECONSTRUCTED
- HYPOTHETICAL
- Mixed flow legend
- Grayscale/print
- Unknown amount
- Range amount

EvidenceCitation

- Page citation
- Paragraph citation
- Region citation
- Derivative citation
- Restricted evidence
- Missing parent warning
- Very long title

EntitySummaryCard

- Person
- Company
- Organization
- Wallet
- Disputed identity
- Unresolved identity
- Long aliases
- Restricted identifiers

EvidenceBalanceMatrix

- Balanced
- Support-heavy
- Contradiction-heavy
- Unknown-heavy
- No evidence
- Keyboard navigation

ReviewDecisionPanel

- Pending
- Changes requested
- Approved
- Rejected
- Self-approval blocked

- Saving
- Permission denied

11. Testing Strategy

Test layer	Tooling expectation	Scope
Type/static	TypeScript + lint	Public props, enum exhaustiveness, import boundaries.
Unit	Vitest/Jest equivalent	Pure rendering logic, formatting, variant mapping.
Interaction	Storybook test/play + Testing Library	Keyboard, click, validation, dialogs, menus, async states.
Accessibility	axe integration + manual keyboard/screen-reader checks	P0 shared components and critical compound patterns.
Visual regression	Chromatic/Percy/self-hosted equivalent	Stable stories at reference viewports, especially analytical semantics.
Integration	Application test suite	Data binding, permissions, routing, server-side authorization.
E2E	Playwright/equivalent	Critical user journeys across composed screens.

CISB-040 — Visual regression approval SHALL require semantic review when a change affects analytical-state, confidence, provenance, classification, or value-flow presentation.

CISB-041 — A pixel-diff approval SHALL NOT override a normative semantic violation.

12. Storybook Viewport Baseline

Viewport	Reference	Use
Desktop review	1440 × 900	Primary high-fidelity composition and data-density review.
Laptop	1280 × 800	Constrained desktop and common field laptop.
Tablet landscape	1024 × 768	Collapsed secondary rails and adaptive analytical workspaces.
Narrow	768 px wide	Stacking behavior and essential task preservation.
Component intrinsic	Auto/min-content/max-content	Atomic component resilience to label/data length.

13. Story Documentation Requirements

- Purpose and when to use / when not to use.
- Public props and semantic variants.
- Accessibility notes and keyboard behavior.
- Content rules and prohibited uses.
- Relevant normative references (UX/UI/IA/High-Fidelity/Screen IDs).
- Known limitations.
- Examples of correct and incorrect semantic usage for high-risk analytical components.

14. Critical Screen-to-Component Traceability

Screen ID	Screen	Mandatory shared component baseline
SCR-EVD-003	Evidence Reader	CMP-003, ANA-004, ANA-005, ANA-012, ANA-023
SCR-ENT-003	Entity Detail	CMP-003, CMP-002, ANA-008, ANA-009, ANA-010
SCR-ENT-005	Entity Match Compare	CMP-006, ANA-008, ANA-004, ANA-001
SCR-TIM-001	Timeline	CMP-005, ANA-016, ANA-011, ANA-012
SCR-VAL-001	Value Flow Workspace	CMP-005, ANA-003, ANA-014, ANA-015, GEN-012
SCR-TYP-002	Typology Match	CMP-005, ANA-020, ANA-004, ANA-019
SCR-HYP-001	Hypothesis Workspace	CMP-005, ANA-017, ANA-018, ANA-019
SCR-ASM-001	Assessment Draft	CMP-004, ANA-021, ANA-002, ANA-019
SCR-GRF-001	Investigation Graph	CMP-005, ANA-013, ANA-014, ANA-015, ANA-012
SCR-PRD-003	Product Editor	CMP-004, ANA-004, ANA-001, ANA-021, ANA-023
SCR-REV-002	Peer Review	CMP-007, ANA-022, ANA-021, ANA-004, ANA-023
SCR-DIS-002	Export Package Builder	CMP-005, ANA-023, ANA-004, GEN-008, GEN-017

CISB-050 — Critical screens SHALL reference shared component IDs in frontend implementation notes or equivalent engineering documentation.

15. Reference Repository Structure

frontend/
 src/
 design-system/
 foundations/
 components/
 general/
 analytical/
 patterns/
 tokens/
 icons/
 index.ts
 features/
 cases/
 evidence/
 entities/
 analysis/
 products/
 administration/
 .storybook/
 main.ts
 preview.ts
 manager.ts
 stories/ (optional docs-only compositions)
 tests/

CISB-060 — Feature packages SHALL NOT deep-import private component implementation files; only public design-system exports are supported contracts.

16. Component Lifecycle and Governance

State	Meaning
Proposed	Need identified; API not stable.
Experimental	Usable in limited feature work; change without migration guarantee.
Stable	Approved public API and Storybook/test coverage.
Deprecated	Replacement exists; migration guidance required.
Retired	Removed from public exports after approved migration window.

CISB-070 — Breaking changes to Stable component APIs SHALL require migration notes and release-note entry.

CISB-071 — Changes to analytical semantic components SHALL be reviewed by UX/design-system owner and a domain/framework owner.

CISB-072 — A component SHOULD be promoted to shared inventory when two or more screens need the same semantic behavior, or immediately when the behavior is safety/security critical.

17. CI/CD and Storybook Publication

- Build Storybook on every pull request affecting frontend/design-system code.
- Run typecheck, lint, unit/interaction tests, and automated accessibility checks.
- Publish ephemeral preview for reviewer access.
- Run visual regression against approved baseline for changed stories.
- Block merge on failed mandatory checks unless a documented exception is approved.
- Publish versioned Storybook for release candidates; production deployment need not expose it publicly.

18. Security and Privacy in Storybook

CISB-080 — Storybook fixtures SHALL contain only synthetic or explicitly approved non-sensitive data.

CISB-081 — Storybook builds SHALL NOT embed production API tokens, production endpoints requiring credentials, or real protected-source identities.

CISB-082 — Authorization behavior MAY be simulated through safe decorators/fixtures, but Storybook SHALL NOT be treated as proof of server-side authorization.

CISB-083 — Private deployment or access control SHOULD be used when Storybook documentation contains internal security architecture details.

19. Component Definition of Done

- ☐ Inventory ID and semantic owner assigned.
- ☐ Public TypeScript API documented.
- ☐ Uses semantic tokens and approved icons.
- ☐ Default + all required variants implemented.
- ☐ Unknown/empty/error/loading/readonly/restricted states implemented where applicable.
- ☐ Keyboard behavior verified.
- ☐ Automated accessibility check passes for supported state set.
- ☐ Storybook stories complete.
- ☐ Interaction tests present for meaningful behaviors.
- ☐ Visual-regression baseline approved for P0/critical analytical components.
- ☐ No sensitive fixture data.
- ☐ Relevant screen IDs and normative references documented.
- ☐ No duplicate local implementation exists in target feature module.

20. Storybook Release Gate

MVP Storybook gate

Before MVP UI freeze, every P0 shared component used by the critical end-to-end scenario SHALL have an approved Storybook entry, required state coverage, accessibility checks, and visual-regression baseline. Missing Storybook coverage for a critical analytical semantic component is a release-quality defect, not merely documentation debt.

Annex A — Minimum Story Coverage by Priority

Priority	Minimum
P0	Default, all variants, key interaction, error/unknown/restricted where applicable, accessibility, critical viewport, visual baseline.
P1	Default, all public variants, basic accessibility, key interaction; visual baseline recommended.
P2	Default and public variants; additional test/story coverage before promotion to production-critical use.

Annex B — Story Naming Convention

Canonical format

`<Group>/<Subgroup>/<Component>` for title; stories use clear state names such as `Direct`, `Reconstructed`, `RestrictedEvidence`, `KeyboardNavigation`, or `LongContent`. Avoid `Example1`, `Test`, and ambiguous names.

Annex C — Pull Request Review Checklist

- ☐ Component belongs in shared library rather than local feature code.
- ☐ No normative semantic is implemented with raw color alone.
- ☐ Public props are typed and minimal.
- ☐ Story set covers non-happy-path states.
- ☐ Keyboard and focus behavior checked.
- ☐ axe violations resolved or documented with approved rationale.
- ☐ Synthetic fixtures only.
- ☐ Visual diffs reviewed semantically.
- ☐ Screen traceability updated if usage changes.
- ☐ Breaking API changes include migration notes.